# worn

An AI-powered digital closet. A case study.

## the problem

I have a lot of clothes I don't wear. Not because I don't like them, but because I forget I own them. I'll be packing for a weekend and reach for the same five tops while a dozen "I love this" pieces sit untouched in the back of my closet. I'm not alone in this. The average person wears about 20% of their wardrobe regularly, and "I have nothing to wear" usually means "I can't remember what I have."

Existing closet apps treat this as a *cataloging* problem. They're databases of your clothes. But the real problem isn't that I lack a list. It's that I lack a *prompt*. I need something to surface forgotten pieces at the moment I'm getting dressed, in a way that actually maps to how I think about outfits ("warm-night dinner date," not "blouse, polyester, $58").

worn is what I'm building to solve that.

## the product

worn is an AI-powered digital closet for iOS. You upload photos of your clothes, and Claude's vision API auto-tags each item across category, color, formality, occasion, weather, and aesthetic. You can then search your closet using natural language. Queries like "sorority formal warm night," "frat party memorial weekend," or "rainy day brunch" return pieces that fit the vibe, *biased toward items you've worn least often*. The whole product hinges on that last part: surfacing what you've forgotten, not just what fits the filters.

The app also tracks how often you wear each item and when you last wore it, so over time worn becomes a record of how your closet actually lives, not just what's in it.

## what's built

- **Native iOS app** in Swift, SwiftUI, and SwiftData
- **Photo upload** from photo library, camera, or Files
- **AI tagging** via Claude vision API, returning structured JSON across six tag dimensions
- **Wear tracking** with count and last-worn date per item

- **Vibe search** that takes a natural-language query and returns ranked items with reasoning, prioritizing under-worn pieces
- **Manual edit** via a pencil-icon inline editor for when the AI gets a name wrong
- **Delete** via long-press context menu or detail-view button
- **Custom design system** with a color palette, typography, and brand identity built around the worn logo

## what’s next

- **Friends and borrowing.** Mark pieces as “open to borrow” and invite specific friends to view your closet. They can request items with a date, time, and optional note, and you approve or deny. The point is closed-circle, opt-in clothes sharing. Not a marketplace, not public, just an extension of how people already share with their close friends.
- **Background removal** for cleaner catalog-style photos. Currently working via Apple’s Vision framework on simple backgrounds; flat-lays on busy fabric remain a known limitation.
- **Re-tag button** for when prompt tuning updates the tagging logic and existing items need refresh.
- **Wear streaks and forgotten-piece nudges** that surface items you haven’t worn in 30+ days as gentle reminders.
- **Android expansion** in Kotlin.

## design ethos

A few principles I keep coming back to as I build:

### 1. Surface what's forgotten, not what's optimized.

The whole reason worn exists is to fight wardrobe blindness. Every ranking decision biases toward under-worn pieces. The user can always sort differently, but the default behavior should always push them toward something they've forgotten.

### 2. The AI suggests pieces; the user builds outfits.

I considered having worn construct full outfits (top + bottom + shoes) and decided against it. People want to see options and mix and match themselves. That's part of the joy of getting dressed. The AI's job is to *narrow* the search to a handful of vibe-appropriate pieces. The human's job is to pick.

### 3. Tags should reflect how people actually think.

Vibe searches like “warm-night dinner date” are really event + weather + time-of-day combos. The tag schema (category, color, formality, vibe, occasion, weather) is designed to match the language users actually use when planning outfits, not industry-standard fashion taxonomies.

### 4. Sustainable as a side effect.

worn isn't a sustainability app. But by making the clothes you already own more visible, it nudges you toward rewearing instead of buying new. That's a quiet win, not a marketing pitch.

### 5. Build the smallest version that's actually useful.

Friends and borrowing are exciting but scoped to v2. The MVP is closet, AI tagging, and vibe search. If that core loop doesn't earn daily use, no amount of social features will fix it.

## tech stack

- **iOS:** Swift, SwiftUI, SwiftData
- **AI:** Anthropic Claude API (Sonnet) for vision tagging and natural-language search
- **Image processing:** Apple Vision framework for background removal
- **Architecture:** Local-first (no backend yet), API calls direct from device

## development context

worn is a self-taught Swift project. I started learning iOS development with this app, using Claude heavily as a real-time tutor while building. I focused on understanding architectural decisions rather than memorizing syntax. The product decisions, what to scope in, what to cut, how to tune AI prompts, and what the data model should capture, are mine.

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